

40009 34

Haskell L-Systems

Submitters

kw1425

Emarking

```
1: Final Tests: Summary for kw1425 of c1
2: PPT 21
3: -----
4:
5:   Public Tests:
6:     student-tests/lsystems-test/lsystems/angle:      4 / 4
7:     student-tests/lsystems-test/lsystems/axiom:      4 / 4
8:     student-tests/lsystems-test/lsystems/rules:      2 / 2
9:     student-tests/lsystems-test/lsystems/lookupChar: 4 / 4
10:    student-tests/lsystems-test/lsystems/expandOne:   2 / 2
11:    student-tests/lsystems-test/lsystems/expand:      6 / 6
12:    student-tests/lsystems-test/lsystems/parse:       3 / 3
13:    student-tests/lsystems-test/lsystems/expandLSystem: 3 / 3
14:    student-tests/lsystems-test/lsystems/move:        7 / 7
15:    student-tests/lsystems-test/lsystems/trace1:      4 / 4
16:    student-tests/lsystems-test/lsystems/trace2:      4 / 4
17:    original-tests/lsystems-test/lsystems/angle:      4 / 4
18:    original-tests/lsystems-test/lsystems/axiom:      4 / 4
19:    original-tests/lsystems-test/lsystems/rules:      2 / 2
20:    original-tests/lsystems-test/lsystems/lookupChar: 4 / 4
21:    original-tests/lsystems-test/lsystems/expandOne:   2 / 2
22:    original-tests/lsystems-test/lsystems/expand:      6 / 6
23:    original-tests/lsystems-test/lsystems/parse:       3 / 3
24:    original-tests/lsystems-test/lsystems/expandLSystem: 3 / 3
25:    original-tests/lsystems-test/lsystems/move:        7 / 7
26:    original-tests/lsystems-test/lsystems/trace1:      4 / 4
27:    original-tests/lsystems-test/lsystems/trace2:      4 / 4
28:
29: Git Repo: git@gitlab.doc.ic.ac.uk:lab2526_autumn/haskell11systems_kw1425.git
30: Commit ID: 66cfc
31: Git Diff: https://gitlab.doc.ic.ac.uk/lab2526_autumn/haskell11systems_kw1425/-/compare/master..66cfc?from_project_id=145289
```

```

1: {-# LANGUAGE StandaloneDeriving #-}
2: import Test.Tasty ( defaultMain, TestTree, testGroup
3:                     , after, DependencyType(AllSucceed) -- used to control tests
4:                     )
5: import Test.Tasty.HUnit (testCase, Assertion)
6: import IC.Approx
7: import IC.Exact
8:
9: import IC.Colour
10: import LSystems
11: import Examples
12:
13: main :: IO ()
14: main = defaultMain tests
15:
16: -- TODO: parse tests, reorder tests
17: -- Some of the later tests depend on 'angle', 'axiom' and 'rules' heavily:
18: -- go implement them first, or the tests won't run!
19: tests :: TestTree
20: tests = testGroup "lsystems"
21:   [ testGroup "angle" (numberedTests angleTests)
22:   , testGroup "axiom" (numberedTests axiomTests)
23:   , testGroup "rules" (numberedTests rulesTests)
24:   , after AllSucceed "/rules/" $
25:     testGroup "lookupChar" (numberedTests lookupCharTests)
26:   , after AllSucceed "/rules/ || /axiom/" $
27:     testGroup "expandOne" (numberedTests expandOneTests)
28:   , after AllSucceed "/rules/ || /axiom/" $
29:     testGroup "expand" (numberedTests expandTests)
30:   , after AllSucceed "/lookupChar/" $
31:     testGroup "parse" (numberedTests parseTests)
32:   , after AllSucceed "/parse/" $
33:     testGroup "expandLSystem" (numberedTests expandLSystemTests)
34:   , testGroup "move" (numberedTests moveTests)
35:   , after AllSucceed "/angle/" $
36:     testGroup "tracel" (numberedTests (traceTests tracel))
37:   , after AllSucceed "/angle/" $
38:     testGroup "trace2" (numberedTests (traceTests trace2))
39:   ]
40:
41: angleTests :: [Assertion]
42: angleTests = [ angle cross --> 90
43:               , angle (LSystem 1 "" []) --> 1
44:               , angle triangle --> 90
45:               , angle arrowHead --> 60
46:               ]
47:
48: axiomTests :: [Assertion]
49: axiomTests = [ axiom (LSystem 0 "+" []) --> "+"
50:               , axiom cross --> "M-M-M-M"
51:               , axiom triangle --> "M"
52:               , axiom arrowHead --> "N"
53:               ]
54:
55: rulesTests :: [Assertion]
56: rulesTests = [ rules cross --> [ ('M', "M-M+M+MM-M-M+M")
57:                                   , ('+', "+")
58:                                   , ('-', "-")
59:                                   ]
60:               , rules (LSystem 0 "" [('M', "N")]) --> [ ('M', "N") ]
61:               ]
62:
63: lookupCharTests :: [Assertion]
64: lookupCharTests =
65:   [ lookupChar [('X', "Yes"), ('Y', "No")] 'X' --> "Yes"
66:   , lookupChar [('Y', "No"), ('X', "Yes")] 'X' --> "Yes"

```

```

67:   , lookupChar (rules peanoGosper) 'M' --> "M+N+N-M-M-M-N+"
68:   , lookupChar (rules triangle) '+' --> "+"
69:   ]
70:
71: expandOneTests :: [Assertion]
72: expandOneTests =
73:   [ expandOne (rules triangle) (axiom triangle) --> "M-M-M-M+M"
74:   , expandOne [('A', "B")] "A" --> "B"
75:   ]
76:
77: expandTests :: [Assertion]
78: expandTests =
79:   [ expand (rules arrowHead) (axiom arrowHead) 2 --> "N+M+N-M-N-M-N+M+N"
80:   , expand (rules dragon) (axiom dragon) 0 --> "MX"
81:   , expand (rules dragon) (axiom dragon) 1 --> "A+MX--MY+"
82:   , expand (rules dragon) (axiom dragon) 5
83:   , -- the backslashes here are called "string gaps", allowing for writing
84:     -- a string across multiple lines
85:     --> "A+A+A+A+MX--MY+--A-MX++MY+---A-A+MX--MY+++A-MX++MY---\
86:         \--A-A+A+MX--MY+--A-MX++MY+---A-A+MX--MY+++A-MX++MY---\
87:         \-A-A+A+A+MX--MY+--A-MX++MY+---A-A+MX--MY+++A-MX++MY+---\
88:         \+A-A+A+MX--MY+--A-MX++MY+---A-A+MX--MY+++A-MX++MY+---+"
89:   ]
90:
91:   , expand (rules tree) (axiom tree) 2
92:   , expand (rules bush) (axiom bush) 3
93:   , --> "MMMM-[MM-[M-[X]+X]+M+MX]-X]+M-[X]+X]+M+MX]-X]+MM[\
94:         \+MMM-[X]+X]+M+MX]-X]-M-[X]+X]+M+MX]-X]+MM-[M-[X]\
95:         \+X]+M+MX]-X]+M-[X]+X]+M+MX]-X]+MM+MMM-[X]+X]+M+M\
96:         \X]-X]-M-[X]+X]+M+MX]-X]+MMMM[+MMMMMM-[M-[X]+X]+M+M\
97:         \MX]-X]+M-[X]+X]+M+MX]-X]+MM+MMM-[X]+X]+M+MX]-X]-M\
98:         \-[X]+X]+M+MX]-X]-MM-[M-[X]+X]+M+MX]-X]+M-[X]+X]+M\
99:         \M+MX]-X]+MM+MMM-[X]+X]+M+MX]-X]-M-[X]+X]+M+MX]-X"
100:   ]
101:
102: parseTests :: [Assertion]
103: parseTests = [ parse "NN[-N[-M][+M][NM]][+N[-M][+M][NM]][NNN[-M][+M][NM]]"
104:               --> [ F, F
105:                   , B [R, F, B [R, F], B [L, F], B [F, F]]
106:                   , B [L, F, B [R, F], B [L, F], B [F, F]]
107:                   , B [F, F, F, B [R, F], B [L, F], B [F, F]]
108:                   ]
109:               , parse "N+M+N-M-N-M-N+M+N"
110:               --> [F, L, F, L, F, R, F, R, F, R, F, L, F, L, F]
111:               , parse "A+MX--MY+"
112:               --> [L, F, R, R, F, L]
113:               ]
114:
115: expandLSystemTests :: [Assertion]
116: expandLSystemTests =
117:   [ expandLSystem tree 2
118:   --> [ F, F
119:       , B [R, F, B [R, F], B [L, F], B [F, F]]
120:       , B [L, F, B [R, F], B [L, F], B [F, F]]
121:       , B [F, F, F, B [R, F], B [L, F], B [F, F]]
122:       ]
123:   , expandLSystem arrowHead 2
124:   --> [F, L, F, L, F, R, F, R, F, R, F, L, F, L, F]
125:   , expandLSystem dragon 1
126:   --> [L, F, R, R, F, L]
127:   ]
128:
129: moveTests :: [Assertion]
130: moveTests = [ move L 90 ((100, 100), 90) --> ((100.0,100.0),180.0)
131:             , move F 60 ((50, 50), 60) --> ((50.5,50.866024),60.0)
132:             , move F 45 ((-25, 180), 180) --> ((-26.0,180.0),180.0)

```

```

Final Tests                                Tests.hs: 3/3                                Katie Wan(kw1425):c1:21

133:      , move R 90 ((200, 200), 180) ~~> ((200.0, 200.0),90.0)
134:      , move R 45 ((-20, 50), 0)    ~~> ((-20.0, 50.0),-45.0)
135:      , move R 90 ((100, 100), 90)   ~~> ((100.0,100.0),0.0)
136:      , move R 30 ((100, 100), 90)   ~~> ((100.0,100.0),60.0)
137:    ]
138:
139: traceTests :: ([Command] -> Float -> Colour -> [ColouredLine]) -> [Assertion]
140: traceTests trace
141:   = [ trace (expandLSystem triangle 1) (angle triangle) Blue
142:     ~~> [ ((0.0,0.0),(1.0,0.0),Blue)
143:           , ((1.0,0.0),(0.99999994,1.0),Blue)
144:           , ((0.99999994,1.0),(2.0,1.0),Blue)
145:           , ((2.0,1.0),(2.0,0.0),Blue)
146:           , ((2.0,0.0),(3.0,0.0),Blue)
147:         ]
148:     , trace (expandLSystem tree 1) (angle tree) Red
149:     ~~> [ ((0.0,0.0),(-4.371139e-8,1.0),Red)
150:           , ((-4.371139e-8,1.0),(0.7071067,1.7071068),Red)
151:           , ((-4.371139e-8,1.0),(-0.7071068,1.7071068),Red)
152:           , ((-4.371139e-8,1.0),(-8.742278e-8,2.0),Red)
153:           , ((-8.742278e-8,2.0),(-1.3113416e-7,3.0),Red)
154:         ]
155:     , trace (expandLSystem galaxy 1) (angle galaxy) Blue
156:     ~~> [ ((0.0,0.0),(-0.5877852,0.809017),Blue)
157:           , ((-0.5877852,0.809017),(0.0,1.6180341),Blue)
158:           , ((0.0,0.0),(-0.9510564,-0.3090172),Blue)
159:           , ((0.0,1.6180341),(0.5877852,0.80901706),Blue)
160:           , ((-0.9510564,-0.3090172),(-1.5388416,0.4999998),Blue)
161:           , ((-1.5388416,0.4999998),(-0.58778507,0.8090168),Blue)
162:           , ((0.0,0.0),(1.1924881e-8,-1.0),Blue)
163:           , ((1.1924881e-8,-1.0),(-0.9510564,-1.3090172),Blue)
164:           , ((-0.9510564,-1.3090172),(-0.9510565,-0.30901718),Blue)
165:           , ((0.0,0.0),(0.95105654,-0.30901694),Blue)
166:           , ((0.95105654,-0.30901694),(0.95105654,-1.309017),Blue)
167:           , ((0.95105654,-1.309017),(5.9604645e-8,-0.9999999),Blue)
168:           , ((0.0,0.0),(0.5877852,0.80901706),Blue)
169:           , ((0.5877852,0.80901706),(1.5388417,0.5000001),Blue)
170:           , ((1.5388417,0.5000001),(0.9510563,-0.3090167),Blue)
171:         ]
172:     , trace (expandLSystem canopy 1) (angle canopy) Blue
173:     ~~> [ ((0.0,0.0),(-4.371139e-8,1.0),Blue)
174:           , ((-4.371139e-8,1.0),(-0.5000001,1.8660254),Blue)
175:           , ((-0.5000001,1.8660254),(-1.0000002,2.732051),Blue)
176:           , ((-4.371139e-8,1.0),(0.49999994,1.8660254),Blue)
177:           , ((0.49999994,1.8660254),(0.9999999,2.732051),Blue)
178:           , ((-4.371139e-8,1.0),(-8.742278e-8,2.0),Blue)
179:           , ((-8.742278e-8,2.0),(0.49999988,2.8660254),Blue)
180:           , ((-8.742278e-8,2.0),(-0.5000001,2.8660254),Blue)
181:           , ((-8.742278e-8,2.0),(-1.3113416e-7,3.0),Blue)
182:         ]
183:   ]
184:
185: -----
186: -- HELPERS
187: {-|
188:   This function just matches up a bunch of assertions to a numerical naming
189:   system, allowing us to distinguish them.
190:
191:   If we wanted, we could provide descriptions to them instead...
192: -|}
193: numberedTests :: [Assertion] -> [TestTree]
194: numberedTests = zipWith (\n -> testCase ("#" ++ show n)) [1..] :: [Integer]
195:
196: deriving instance Eq Command

```

Final Tests**Bench.hs: 1/1****Katie Wan(kw1425):c1:21**

```
1: {-# LANGUAGE DeriveGeneric, DeriveAnyClass, StandaloneDeriving #-}
2: import Test.Tasty.Bench ( defaultMain, Benchmark, bgroup, bench, nf, env )
3: import GHC.Generics
4: import Control.DeepSeq
5:
6: import IC.Colour
7: import LSystems
8: import Examples
9:
10: main :: IO ()
11: main = defaultMain [benchmarks]
12:
13: benchmarks :: Benchmark
14: benchmarks = env @[Command], Float, Colour) (pure (expandLSystem tree 5, angle ↗
tree, Green)) (\ ~(cmds, ang, col) ->
15:   bgroup "lsystems" [ bench "trace1" (nf (trace1 cmds ang) col)
16:                     , bench "trace2" (nf (trace2 cmds ang) col)
17:                     ])
18:
19: -- 'ere be Dragons
20: -- This stuff is used to allow for the strict evaluation of Commands and Colours
21: -- Don't worry about how this works :)
22: deriving instance Generic Command
23: deriving instance NFData Command
24: deriving instance Generic Colour
25: deriving instance NFData Colour
```

2 / 2

Very nice

Final Tests

LSystems.hs: 1/3

Katie Wan(kw1425):c1:21

```

1: module LSystems ( LSystem(LSystem), ColouredLine, Command(..)
2:   , angle, axiom, rules, lookupChar
3:   , expandOne, expand, move, parse, trace1, trace2
4:   , expandLSystem ) where
5:
6: import IC.Colour
7: import Data.Maybe (fromMaybe)
8: import Data.List (foldl')
9:
10: -- Part 1 - Warmup
11: -----
12: type Rules a = [(Char, [a])]
13: data LSystem = LSystem Float [Char] (Rules Char)
14:
15: -- Functions for working with systems. Skeleton optional...
16:
17: -- Returns the rotation angle for the given system.
18: angle :: LSystem -> Float
19: angle (LSystem angle _) = angle
20:
21: -- Returns the axiom string for the given system.
22: axiom :: LSystem -> [Char]
23: axiom (LSystem _ axiom _) = axiom
24:
25: -- Returns the set of rules for the given system.
26: rules :: LSystem -> Rules Char
27: rules (LSystem _ _ rules) = rules
28:
29: --
30: -- Pre: the character has a binding in the Rules list
31: --
32: lookupChar :: Rules a -> Char -> [a]
33: lookupChar (r:rs) c
34:   | char == c = value
35:   | otherwise = lookupChar rs c
36: where (char, value) = r
37:
38: -- using lookup from prelude (idk if this is cheating or not)
39: lookupChar :: Rules a -> Char -> [a]
40: lookupChar rules c = fromMaybe [] (lookup c rules)
41:
42: -- using filter
43: lookupChar :: Rules a -> Char -> [a]
44: lookupChar rules c = concatMap snd (filter ((== c) . fst) rules)
45:
46: -- i don't think this is the best way to select the second value from a
47: -- list of pairs when there is only 1 pair in the list, but I can't think
48: -- of another way to do this without using head
49:
50: -- using foldr
51: lookupChar :: Rules a -> Char -> [a]
52: lookupChar rules c = foldr findKey [] rules
53:   where
54:     findKey :: (Char, [a]) -> [a] -> [a]
55:     findKey (ch, val) acc
56:       | ch == c = val
57:       | otherwise = acc
58:
59:
60: -- Part 2: Expansion and Parsing
61: type Parser a = String -> (a, String)
62: data Command = F | L | R | B [Command] deriving Show
63:
64: --
65: -- Expand command string s once using rule table r
66: --

```

instead of defining 3 different functions,
you can alter the data type to use
record syntax:

```

data LSystem = LSystem {
  angle :: Float
, axiom :: [Char]
, rules :: Rules Char }

```

very good using
fromMaybe here

↳ here, using head would be good enough.

IF the resulting list was empty, something
has already gone wrong.

Just avoid using head on things that could be
empty, or where pattern matching is possible

Final Tests

LSystems.hs: 2/3

Katie Wan(kw1425):c1:21

```

67: expandOne :: Rules Char -> [Char] -> [Char]
68: -- expandOne r (x:xs) = lookupChar r x ++ expandOne r xs -- this is the shape of /
a concatMap
69: expandOne r = concatMap (lookupChar r)
70:
71: --
72: -- Expand command string s n times using rule table r
73: --
74: expand :: Rules Char -> [Char] -> Int -> [Char]
75: expand _ x 0 = x
76: expand r x n = expand r newX (n-1)
77:   where newX = expandOne r x
78:
79: parse :: [Char] -> [Command]
80: parse = fst.parse'
81:   where
82:     parse' :: Parser [Command]
83:     parse' [] = ([], [])
84:     parse' (x:xs)
85:       | x == '[' = ([], xs)
86:       | x == '[' = (B inner : outer, rem2)
87:       | otherwise = (cmd ++ inner, rem1)
88:     where
89:       (inner, rem1) = parse' xs
90:       (outer, rem2) = parse' rem1
91:       cmd = lookupChar commandMap x
92:
93: expandLSystem :: LSystem -> Int -> [Command]
94: expandLSystem (LSystem _ a r) = parse.expand r a
95:
96: -- Part 3: Turtles
97: type Vertex = (Float, Float)
98: type TurtleState = (Vertex, Float)
99: type ColouredLine = (Vertex, Vertex, Colour)
100:
101: initialState :: TurtleState
102: initialState = ((0, 0), 90)
103:
104: -- Move a turtle.
105: --
106: -- F moves distance 1 in the current direction.
107: -- L rotates left according to the given angle.
108: -- R rotates right according to the given angle.
109: move :: Command -> Float -> TurtleState -> TurtleState
110: move L angle (l, o) = (l, o+angle)
111: move R angle (l, o) = (l, o-angle)
112: move F _ ((x, y), o) = ((x + dx, y + dy), o)
113:   where
114:     oRad = degreesToRadians o
115:     dx = cos oRad
116:     dy = sin oRad
117:
118:
119: trace1 :: [Command] -> Float -> Colour -> [ColouredLine]
120: trace1 cmds a col = go cmds initialState
121:   where
122:     go :: [Command] -> TurtleState -> [ColouredLine]
123:     go [] _ = []
124:     go (F:cs) st = (pos1, pos2, col) : go cs st'
125:       where st' = move F a st
126:             pos1 = fst st
127:             pos2 = fst st'
128:     go (B cs' : cs) st = inner ++ outer
129:       where
130:         inner = go cs' st
131:         outer = go cs st

```

Try using higher-order functions to
avoid recursion.
(fold, iterate)

4.5/5

you can use pattern matching or
case x of instead of guards

1/1

↳ good on using composition here

3/3

2/3

→ you can use @ and pattern matching instead
of using fst

```

132: go (c:cs) st = go cs (move c a st)
133:
134: -- This version uses an explicit stack of residual commands and turtle states
135: trace2 :: [Command] -> Float -> Colour -> [ColouredLine]
136: trace2 cmds a col = go cmds initialState []
137: where
138:   go :: [Command] -> TurtleState -> [[Command], TurtleState] -> ✓
[ColouredLine]
139:   go [] _ [] = []
140:   go [] _ ((cs, state):stack') = go cs state stack'
141:
142:   go (F:cs) state stack = (pos1, pos2, col) : go cs state' stack
143:   where state' = move F a state
144:         pos1 = fst state
145:         pos2 = fst state' → same advice here
146:
147:   go (B cs' : cs) state stack = go cs' state ((cs, state) : stack)
148:
149:   go (c:cs) state stack = go cs (move c a state) stack
150:
151:
152: -- results for benchmarks
153: {-
154: trace1: OK
155:      43.6 fms ± 1.6 fms
156: trace2: OK
157:      34.1 fms ± 1.5 fms
158: -} ✓
159:
160: -- trace1 is slower than trace2 as expected, as the stack in trace2 allows for ✓
less recursive calls.
161:
162:
163: -- Provided Helper Functions
164: -----
165:
166: degreesToRadians :: Float -> Float
167: degreesToRadians x = (x / 180) * pi
168:
169: commandMap :: Rules Command
170: commandMap = [ ('M', [F])
171:              , ('N', [F])
172:              , ('X', [])
173:              , ('Y', [])
174:              , ('A', [])
175:              , ('+', [L])
176:              , ('-', [R])
177:              ]

```

2/3

19.5

30

Final Tests

testResults.txt: 1/6

Katie Wan(kw1425):c1:21

```
1: ----- Test Output -----
2: copying lsystems.cabal from skeleton
3: copying cabal.project from skeleton
4: Config file path source is default config file.
5: Config file not found: /root/.config/cabal/config
6: Writing default configuration to /root/.config/cabal/config
7: Resolving dependencies...
8: Build profile: -w ghc-9.6.7 -O2
9: In order, the following will be built (use -v for more details):
10: - ObjectName-1.1.0.2 (lib) (requires build)
11: - StateVar-1.2.2 (lib) (requires build)
12: - call-stack-0.4.0 (lib) (requires build)
13: - colour-2.3.6 (lib) (requires build)
14: - fixed-0.3 (lib) (requires build)
15: - half-0.3.3 (lib) (requires build)
16: - loop-0.3.0 (lib) (requires build)
17: - prettyprinter-1.7.1 (lib) (requires build)
18: - primitive-0.9.1.0 (lib) (requires build)
19: - semigroups-0.20 (lib) (requires build)
20: - splitmix-0.1.3.1 (lib) (requires build)
21: - tagged-0.8.9 (lib) (requires build)
22: - vector-stream-0.1.0.1 (lib) (requires build)
23: - ansi-terminal-types-1.1.3 (lib) (requires build)
24: - OpenGLRaw-3.3.4.1 (lib) (requires build)
25: - random-1.3.1 (lib) (requires build)
26: - vector-0.13.2.0 (lib) (requires build)
27: - ansi-terminal-1.1.3 (lib) (requires build)
28: - GLURaw-2.0.0.5 (lib) (requires build)
29: - matrix-0.3.6.4 (lib) (requires build)
30: - prettyprinter-ansi-terminal-1.1.3 (lib) (requires build)
31: - OpenGL-3.0.3.0 (lib) (requires build)
32: - optparse-applicative-0.19.0.0 (lib) (requires build)
33: - GLUT-2.7.0.16 (lib) (requires build)
34: - tasty-1.5.3 (lib) (requires build)
35: - lsystems-0.1.0.0 (lib) (first run)
36: - tasty-hunit-0.10.2 (lib) (requires build)
37: - lsystems-0.1.0.0 (test:lsystems-test) (first run)
38: Starting   StateVar-1.2.2 (lib)
39: Starting   call-stack-0.4.0 (lib)
40: Starting   colour-2.3.6 (lib)
41: Starting   ObjectName-1.1.0.2 (lib)
42: Building   colour-2.3.6 (lib)
43: Building   ObjectName-1.1.0.2 (lib)
44: Building   StateVar-1.2.2 (lib)
45: Building   call-stack-0.4.0 (lib)
46: Installing ObjectName-1.1.0.2 (lib)
47: Installing call-stack-0.4.0 (lib)
48: Installing StateVar-1.2.2 (lib)
49: Completed  ObjectName-1.1.0.2 (lib)
50: Starting   fixed-0.3 (lib)
51: Completed  call-stack-0.4.0 (lib)
52: Starting   half-0.3.3 (lib)
53: Completed  StateVar-1.2.2 (lib)
54: Starting   loop-0.3.0 (lib)
55: Building   fixed-0.3 (lib)
56: Building   loop-0.3.0 (lib)
57: Building   half-0.3.3 (lib)
58: Installing loop-0.3.0 (lib)
59: Installing fixed-0.3 (lib)
60: Completed  loop-0.3.0 (lib)
61: Starting   prettyprinter-1.7.1 (lib)
```


Final Tests

testResults.txt: 2/6

Katie Wan(kw1425):c1:21

```
62: Completed fixed-0.3 (lib)
63: Starting primitive-0.9.1.0 (lib)
64: Building prettyprinter-1.7.1 (lib)
65: Building primitive-0.9.1.0 (lib)
66: Installing colour-2.3.6 (lib)
67: Installing half-0.3.3 (lib)
68: Completed colour-2.3.6 (lib)
69: Starting semigroups-0.20 (lib)
70: Completed half-0.3.3 (lib)
71: Starting splitmix-0.1.3.1 (lib)
72: Building semigroups-0.20 (lib)
73: Building splitmix-0.1.3.1 (lib)
74: Installing semigroups-0.20 (lib)
75: Completed semigroups-0.20 (lib)
76: Starting tagged-0.8.9 (lib)
77: Installing splitmix-0.1.3.1 (lib)
78: Building tagged-0.8.9 (lib)
79: Completed splitmix-0.1.3.1 (lib)
80: Starting vector-stream-0.1.0.1 (lib)
81: Building vector-stream-0.1.0.1 (lib)
82: Installing tagged-0.8.9 (lib)
83: Completed tagged-0.8.9 (lib)
84: Starting ansi-terminal-types-1.1.3 (lib)
85: Building ansi-terminal-types-1.1.3 (lib)
86: Installing vector-stream-0.1.0.1 (lib)
87: Completed vector-stream-0.1.0.1 (lib)
88: Starting OpenGLRaw-3.3.4.1 (lib)
89: Installing primitive-0.9.1.0 (lib)
90: Installing ansi-terminal-types-1.1.3 (lib)
91: Building OpenGLRaw-3.3.4.1 (lib)
92: Completed primitive-0.9.1.0 (lib)
93: Starting random-1.3.1 (lib)
94: Completed ansi-terminal-types-1.1.3 (lib)
95: Starting vector-0.13.2.0 (lib)
96: Building random-1.3.1 (lib)
97: Building vector-0.13.2.0 (lib)
98: Installing prettyprinter-1.7.1 (lib)
99: Completed prettyprinter-1.7.1 (lib)
100: Starting ansi-terminal-1.1.3 (lib)
101: Building ansi-terminal-1.1.3 (lib)
102: Installing ansi-terminal-1.1.3 (lib)
103: Completed ansi-terminal-1.1.3 (lib)
104: Starting prettyprinter-ansi-terminal-1.1.3 (lib)
105: Building prettyprinter-ansi-terminal-1.1.3 (lib)
106: Installing prettyprinter-ansi-terminal-1.1.3 (lib)
107: Completed prettyprinter-ansi-terminal-1.1.3 (lib)
108: Starting optparse-applicative-0.19.0.0 (lib)
109: Building optparse-applicative-0.19.0.0 (lib)
110: Installing optparse-applicative-0.19.0.0 (lib)
111: Completed optparse-applicative-0.19.0.0 (lib)
112: Starting tasty-1.5.3 (lib)
113: Building tasty-1.5.3 (lib)
114: Installing random-1.3.1 (lib)
115: Completed random-1.3.1 (lib)
116: Installing tasty-1.5.3 (lib)
117: Completed tasty-1.5.3 (lib)
118: Starting tasty-hunit-0.10.2 (lib)
119: Building tasty-hunit-0.10.2 (lib)
120: Installing tasty-hunit-0.10.2 (lib)
121: Completed tasty-hunit-0.10.2 (lib)
122: Installing OpenGLRaw-3.3.4.1 (lib)
```

Final Tests**testResults.txt: 3/6****Katie Wan(kw1425):c1:21**

```
123: Completed      OpenGLRaw-3.3.4.1 (lib)
124: Starting        GLURaw-2.0.0.5 (lib)
125: Building        GLURaw-2.0.0.5 (lib)
126: Installing      GLURaw-2.0.0.5 (lib)
127: Completed       GLURaw-2.0.0.5 (lib)
128: Starting        OpenGL-3.0.3.0 (lib)
129: Building        OpenGL-3.0.3.0 (lib)
130: Installing      vector-0.13.2.0 (lib)
131: Completed       vector-0.13.2.0 (lib)
132: Starting        matrix-0.3.6.4 (lib)
133: Building        matrix-0.3.6.4 (lib)
134: Installing      OpenGL-3.0.3.0 (lib)
135: Completed       OpenGL-3.0.3.0 (lib)
136: Starting        GLUT-2.7.0.16 (lib)
137: Building        GLUT-2.7.0.16 (lib)
138: Installing      GLUT-2.7.0.16 (lib)
139: Completed       GLUT-2.7.0.16 (lib)
140: Installing      matrix-0.3.6.4 (lib)
141: Completed       matrix-0.3.6.4 (lib)
142: Configuring library for lsystems-0.1.0.0...
143: Preprocessing library for lsystems-0.1.0.0...
144: Building library for lsystems-0.1.0.0...
145: [1 of 5] Compiling IC.Colour      ( src/IC/Colour.hs, /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/IC/Colour.o, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/IC/Colour.dyn_o )
146: [2 of 5] Compiling IC.Graphics    ( src/IC/Graphics.hs, /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/IC/Graphics.o, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/IC/Graphics.dyn_o )
147: [3 of 5] Compiling LSystems      ( src/LSystems.hs, /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/LSystems.o, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/LSystems.dyn_o )
148: [4 of 5] Compiling Examples      ( src/Examples.hs, /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/Examples.o, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/Examples.dyn_o )
149: [5 of 5] Compiling Renderer      ( src/Renderer.hs, /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/Renderer.o, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/opt/build/Renderer.dyn_o )
150: Configuring test suite 'lsystems-test' for lsystems-0.1.0.0...
151: Preprocessing test suite 'lsystems-test' for lsystems-0.1.0.0...
152: Building test suite 'lsystems-test' for lsystems-0.1.0.0...
153: [1 of 3] Compiling IC.Approx      ( test/IC/Approx.hs, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test-tmp/IC/Approx.o )
154: [2 of 3] Compiling IC.Exact      ( test/IC/Exact.hs, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test-tmp/IC/Exact.o )
155: [3 of 3] Compiling Main          ( test/Tests.hs, ✓
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test-tmp/Main.o )
156: [4 of 4] Linking /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test
157: Build profile: -w ghc-9.6.7 -O2
158: In order, the following will be built (use -v for more details):
159: - lsystems-0.1.0.0 (test:lsystems-test) (ephemeral targets)
160: Preprocessing test suite 'lsystems-test' for lsystems-0.1.0.0...
161: Building test suite 'lsystems-test' for lsystems-0.1.0.0...
162: Running 1 test suites...
163: Test suite lsystems-test: RUNNING...
164: lsystems
165:   angle
166:     #1: OK
167:     #2: OK
168:     #3: OK
169:     #4: OK
170:   axiom
171:     #1: OK
172:     #2: OK
173:     #3: OK
174:     #4: OK
175:   rules
```

Final Tests

testResults.txt: 4/6

Katie Wan(kw1425):c1:21

```
176:      #1: OK
177:      #2: OK
178: lookupChar
179:      #1: OK
180:      #2: OK
181:      #3: OK
182:      #4: OK
183: expandOne
184:      #1: OK
185:      #2: OK
186: expand
187:      #1: OK
188:      #2: OK
189:      #3: OK
190:      #4: OK
191:      #5: OK
192:      #6: OK
193: parse
194:      #1: OK
195:      #2: OK
196:      #3: OK
197: expandLSystem
198:      #1: OK
199:      #2: OK
200:      #3: OK
201: move
202:      #1: OK
203:      #2: OK
204:      #3: OK
205:      #4: OK
206:      #5: OK
207:      #6: OK
208:      #7: OK
209: trace1
210:      #1: OK
211:      #2: OK
212:      #3: OK
213:      #4: OK
214: trace2
215:      #1: OK
216:      #2: OK
217:      #3: OK
218:      #4: OK
219:
220: All 43 tests passed (0.01s)
221:
222: Test suite lsystems-test: PASS
223: Test suite logged to:
224: /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/test/lsystems-0.1.0.0-lsystems-test.log
225: 1 of 1 test suites (1 of 1 test cases) passed.
226: copying test from skeleton
227: Build profile: -w ghc-9.6.7 -O2
228: In order, the following will be built (use -v for more details):
229: - lsystems-0.1.0.0 (test:lsystems-test) (file test/Tests.hs changed)
230: Preprocessing test suite 'lsystems-test' for lsystems-0.1.0.0...
231: Building test suite 'lsystems-test' for lsystems-0.1.0.0...
232: [3 of 3] Compiling Main          ( test/Tests.hs, /
/tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test-tmp/Main.o ) [Source file changed]
233: [4 of 4] Linking /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/build/lsystems-test/lsystems-test [Objects changed]
234: Build profile: -w ghc-9.6.7 -O2
235: In order, the following will be built (use -v for more details):
```

Final Tests

testResults.txt: 5/6

Katie Wan(kw1425):c1:21

```
236: - lsystems-0.1.0.0 (test:lsystems-test) (ephemeral targets)
237: Preprocessing test suite 'lsystems-test' for lsystems-0.1.0.0...
238: Building test suite 'lsystems-test' for lsystems-0.1.0.0...
239: Running 1 test suites...
240: Test suite lsystems-test: RUNNING...
241: lsystems
242:   angle
243:     #1: OK
244:     #2: OK
245:     #3: OK
246:     #4: OK
247:   axiom
248:     #1: OK
249:     #2: OK
250:     #3: OK
251:     #4: OK
252:   rules
253:     #1: OK
254:     #2: OK
255:   lookupChar
256:     #1: OK
257:     #2: OK
258:     #3: OK
259:     #4: OK
260:   expandOne
261:     #1: OK
262:     #2: OK
263:   expand
264:     #1: OK
265:     #2: OK
266:     #3: OK
267:     #4: OK
268:     #5: OK
269:     #6: OK
270:   parse
271:     #1: OK
272:     #2: OK
273:     #3: OK
274:   expandLSystem
275:     #1: OK
276:     #2: OK
277:     #3: OK
278:   move
279:     #1: OK
280:     #2: OK
281:     #3: OK
282:     #4: OK
283:     #5: OK
284:     #6: OK
285:     #7: OK
286:   trace1
287:     #1: OK
288:     #2: OK
289:     #3: OK
290:     #4: OK
291:   trace2
292:     #1: OK
293:     #2: OK
294:     #3: OK
295:     #4: OK
296:
```

Final Tests

testResults.txt: 6/6

Katie Wan(kw1425):c1:21

```
297: All 43 tests passed (0.01s)
298:
299: Test suite lsystems-test: PASS
300: Test suite logged to:
301: /tmp/d20251024-34-catlao/dist-newstyle/build/x86_64-linux/ghc-9.6.7/lsystems-0.1.0.0/t/lsystems-test/opt/test/lsystems-0.1.0.0-lsystems-test.log
302: 1 of 1 test suites (1 of 1 test cases) passed.
303:
304: ----- Test Errors -----
305: Warning: The package list for 'hackage.haskell.org' is 31 days old.
306: Run 'cabal update' to get the latest list of available packages.
```